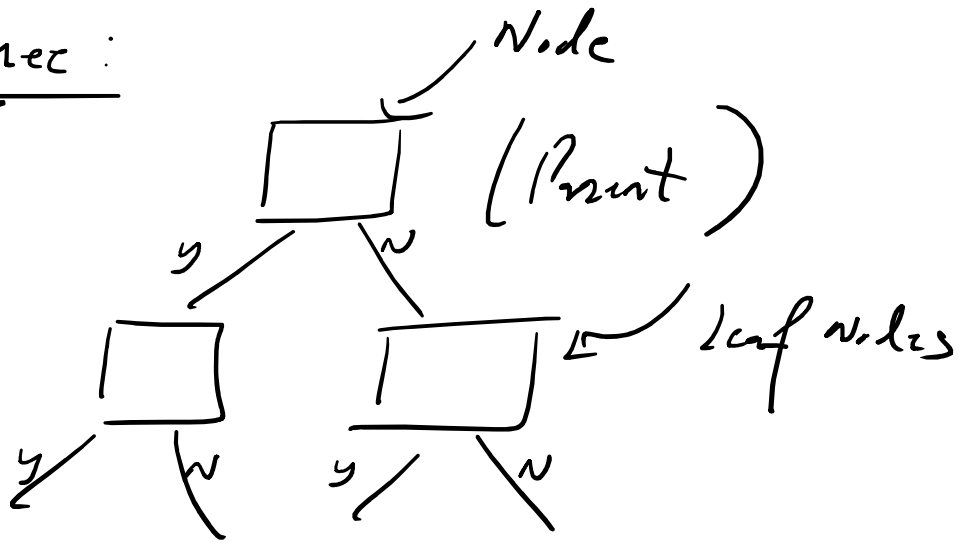
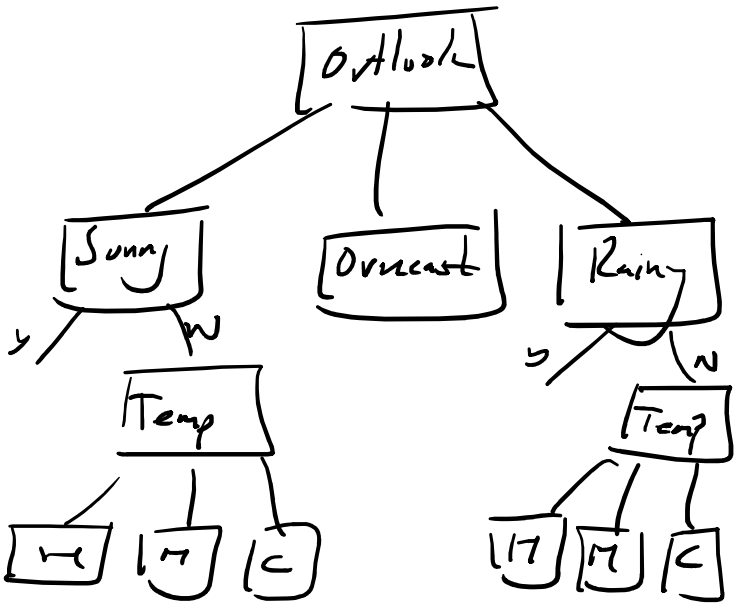
Mod 4.0

12 October 2026 09:07

=> Decision Tree :



Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No



=> Entropy :

↳ Entropy is a measure of purity.

↳ Tells where to split Decision Tree.

↳ Pre-requisite for Information Gain.

↳ Pre-requisite for "Information Gain".

$$\text{Entropy}(\text{Situation}) = \sum_{i=1}^{c} -p_i \log_2 p_i$$

Example:

$$\text{Entropy}(\text{Play Tennis}) = \sum_{i=1}^{c} -p_i \log_2 p_i$$

$$\text{Entropy}(9,5) = \left\{\left(-\frac{9}{14}\log_2\frac{9}{14}\right) + \left(-\frac{5}{14}\log_2\frac{5}{14}\right)\right\}$$

$$= \{0.53 + 0.41\}$$

$$= \boxed{0.94}$$

$$\text{Entropy}(\text{Play Tennis}, \text{Outlook}) = \sum_{i=1}^{c} -p_i \log_2 p_i$$

$$E(\text{PT}, O) = P(\text{Sunny}) \times E(2,3) + P(\text{Overcast}) \times E(4,0) + P(\text{Rainy}) \times E(3,2)$$

$$= \left(\frac{5}{14}\right) \times \left\{ -\frac{2}{5} \log_2 \frac{2}{5} + \left(-\frac{3}{5}\right) \log_2 \frac{3}{5} \right\}$$

$$+ \left(\frac{4}{14}\right) \times \left\{ -\frac{4}{4} \log_2 \frac{4}{4} \right\}$$

$$+ \left(\frac{5}{14}\right) \times \left\{ -\frac{3}{5} \log_2 \frac{3}{5} + \left(-\frac{2}{5}\right) \log_2 \frac{2}{5} \right\}$$

$$= \frac{5}{14} \times 0.971 + \frac{4}{14} \times 0 + \frac{5}{14} \times 0.971$$

$$= 0.693$$

$$\text{Entropy}(PT, \text{Temp}) = P(\text{Hot}) \times E(2,2) + P(\text{Mild}) \times E(4,2) + P(\text{Cool}) \times E(3,1)$$

$$= \frac{4}{14} \times 1 + \frac{6}{14} \times 0.92 \times \frac{4}{14} \times 0.81$$

$$= 0.9114$$

$$\text{Entropy}(PT, \text{Humidity}) = P(\text{High}) \times E(3,4) + P(\text{Normal}) \times E(6,1)$$

$$= 0.7885$$

$$\text{Entropy}[PT, \text{Windy}] = P(\text{Strong}) \times E(3,3) + P(\text{Weak}) \times E(6,2)$$

$$= \boxed{0.892}$$

$\Rightarrow$ Information Gain:

$$\text{Gain}(Y, c) = \text{Entropy}(Y) - \sum_{v \in c} \frac{|Y_v|}{Y} \times \text{Entropy}(Y_v)$$

$\Rightarrow G(PT, \text{Outlook}) = 0.94 - 0.693 = \boxed{0.247}$ 1st

$\Rightarrow G(PT, \text{Temp}) = 0.94 - 0.9114 = \boxed{0.0286}$ 4th

$\Rightarrow G(PT, \text{Humidity}) = 0.94 - 0.7885 = \boxed{0.1515}$ 2nd

$\Rightarrow G(PT, \text{Windy}) = 0.94 - 0.892 = \boxed{0.048}$ 3rd